

ANKENY, IA, USA

WMO: 725466

Lat: 41.691N

Lon: 93.566W

Elev: 910

StdP: 14.22

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04938

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.4	(c) 0.4	(d) -13.3	(e) 2.8	(f) -2.6	(g) -8.5	(h) 3.6	(i) 2.5	(j) 26.6	(k) 23.9	(l) 24.6	(m) 24.7	(n) 7.6	(o) 310	(p) 0.518

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	92.9	75.6	90.2	75.2	87.7	74.0	79.6	88.3	77.6	86.7	75.8	84.4	10.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.1	145.9	85.3	74.7	134.6	83.4	72.8	125.7	81.6	43.9	88.2	41.6	86.5	39.8	84.6	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 23.4	(b) 20.1	(c) 18.0	(e) -12.7	(f) 97.8	(g) 7.2	(h) 3.6	(i) -17.9	(j) 100.4	(k) -22.1	(l) 102.5	(m) -26.2	(n) 104.5	(o) -31.4	(p) 107.2		
(a) 23.4	(b) 20.1	(c) 18.0	(e) -12.7	(f) 82.3	(g) 7.3	(h) 1.9	(i) -18.0	(j) 83.7	(k) -22.2	(l) 84.8	(m) -26.3	(n) 85.9	(o) -31.6	(p) 87.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.6	22.7	26.3	38.9	51.2	61.9	71.3	75.1	72.8	65.2	52.2	39.9	28.2	
	DBStd	20.59	12.76	12.81	12.03	10.00	8.52	6.16	5.76	5.91	8.01	9.08	10.00	10.78	
	HDD50	3091	847	664	382	101	8	0	0	0	3	84	326	676	
	HDD65	6212	1312	1082	811	425	164	15	2	7	93	407	753	1141	
	CDD50	3311	0	2	37	137	375	639	777	707	460	153	23	1	
	CDD65	958	0	0	2	11	66	204	314	250	100	11	0	0	
	CDH74	9478	0	0	23	160	639	1902	3257	2351	1010	132	4	0	
	CDH80	3455	0	0	3	38	181	661	1335	883	328	26	0	0	
Wind	WSAvg	7.8	8.8	9.0	9.3	9.8	8.2	6.9	5.9	5.4	6.2	7.7	8.6	8.3	
Precipitation	PrecAvg	35.1	0.9	1.2	2.1	4.1	5.2	5.1	4.3	3.9	2.9	2.3	1.9	1.3	
	PrecMax	54.3	2.2	2.0	6.3	7.8	9.6	12.9	10.9	11.5	5.6	6.9	5.0	4.8	
	PrecMin	23.0	0.4	0.2	0.0	1.7	1.5	0.9	1.0	0.0	0.8	0.2	0.1	0.0	
	PrecStd	7.6	0.4	0.5	1.4	1.4	2.1	3.1	2.5	2.8	1.5	1.6	1.1	1.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.8	63.7	77.1	85.6	90.6	93.1	98.6	96.5	91.2	83.8	72.8	58.3	
		MCWB	46.1	49.4	59.5	67.4	71.6	77.2	75.0	75.7	72.5	66.6	59.2	49.9	
	2%	DB	49.9	55.9	70.2	78.8	84.3	90.1	93.4	91.2	88.0	77.4	64.7	52.1	
		MCWB	41.8	47.1	57.0	61.6	69.4	75.4	76.6	75.0	72.8	64.2	55.6	46.2	
	5%	DB	44.6	49.7	63.6	73.1	81.0	86.9	90.4	88.3	83.8	72.6	60.7	46.4	
		MCWB	38.7	42.8	54.2	58.6	66.1	73.6	76.4	74.1	70.9	61.5	52.1	41.6	
	10%	DB	38.8	43.6	57.3	69.6	77.0	83.8	87.6	85.4	80.7	68.0	55.2	43.2	
		MCWB	35.0	38.3	48.2	56.2	64.1	72.1	75.3	73.9	68.0	58.7	48.2	38.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.4	52.6	63.9	68.8	75.0	80.3	83.6	80.7	77.1	72.2	61.5	53.2
			MCDB	55.0	61.4	73.3	80.3	86.0	88.9	91.3	90.6	87.1	79.0	69.1	55.5
2%		WB	42.1	47.7	59.6	64.2	71.9	78.0	80.4	78.2	74.9	67.6	57.8	46.6	
		MCDB	48.7	56.5	66.6	74.3	81.6	87.1	88.3	87.5	83.8	73.3	64.3	49.4	
5%		WB	38.8	42.5	54.4	60.9	69.2	75.7	78.2	76.5	72.7	64.0	53.1	42.2	
		MCDB	43.7	49.0	63.2	71.8	77.5	83.9	86.7	85.4	81.0	69.6	59.4	47.3	
10%		WB	35.6	38.7	49.6	57.8	66.9	73.6	76.1	74.5	70.3	59.5	49.3	38.9	
		MCDB	39.3	43.6	57.7	66.4	74.1	81.6	84.6	82.8	77.4	66.1	55.1	42.9	
Mean Daily Temperature Range		5% DB	MDBR	16.6	17.7	19.3	21.5	20.0	19.2	20.2	20.5	22.5	20.7	18.9	15.6
			MCDBR	23.9	25.4	28.7	29.1	24.6	22.0	23.2	24.1	25.6	27.8	26.9	21.8
	MCWBR		17.5	17.0	17.4	16.7	12.0	11.4	10.6	11.2	12.6	15.4	17.8	16.4	
	5% WB	MCDBR	22.6	24.1	26.3	25.3	21.3	19.7	20.0	20.6	20.9	21.5	23.7	20.0	
		MCWBR	17.1	17.1	17.2	16.0	11.7	12.3	11.3	11.4	11.6	15.0	17.3	16.0	
Clear-Sky Solar Irradiance	taub	0.291	0.317	0.330	0.387	0.415	0.426	0.443	0.425	0.390	0.345	0.301	0.292		
	taud	2.414	2.341	2.401	2.262	2.254	2.270	2.207	2.271	2.319	2.441	2.501	2.429		
	Ebn at Noon	273	281	291	280	274	270	264	265	267	268	268	263		
	Edh at Noon	25	31	33	41	43	42	44	41	36	28	23	23		
All-Sky Solar Radiation	RadAvg	608	926	1201	1505	1777	1976	1998	1728	1449	973	661	507		
	RadStd	58	63	114	117	140	187	145	113	106	114	55	47		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page